

Kubernetes Cheat Sheet

Common Options

alias

Set an alias for kubectl.

-o=json

Output format in JSON.

-o=yaml

Output format in YAML.

-o=wide

Output in the plain-text format with any additional information, and for pods, the node name is included.

-n

Alias for namespace.

-f

Filename, directory, or URL to files to use to create a resource.

-l

Filter using a specified label.

-h

Help for kubectl.

Configuration Files

kubectl create -f <configuration file>

Create objects.

kubectl create -f <configuration file directory>

Create objects in all manifest files in a directory.

kubectl create -f <url>

Create objects from a URL.

kubectl delete -f <configuration file>

Delete an object.

Cluster Management and Context

kubectl cluster-info

Display endpoint information about the master and services in the cluster.

kubectl version

Display the Kubernetes version running on the client and server.

kubectl config view

Get the configuration of the cluster.

kubectl config view -o jsonpath='{.users[*].name}'

Get a list of users.

kubectl config current-context

Display the current context.

kubectl config get-contexts

Display a list of contexts.

kubectl config use-context <cluster name>

Set the default context.

kubectl api-resources

List the API resources that are available.

kubectl api-versions

List the API versions that are available.

Daemonsets

kubectl get daemonset

List one or more daemonsets.

kubectl edit daemonset <daemonset_name>

Edit and update the definition of one or more daemonset.

kubectl delete daemonset <daemonset_name>

Delete a daemonset.

kubectl create daemonset <daemonset_name>

Create a new daemonset.

kubectl rollout daemonset

Manage the rollout of a daemonset.

kubectl describe ds <daemonset_name> -n <namespace_name>

Display the detailed state of daemonsets within a namespace.

Deployments

kubectl get deployment

List one or more deployments.

kubectl describe deployment <deployment_name>

Display the detailed state of one or more deployments.

kubectl edit deployment <deployment_name>

Edit and update the definition of one or more deployments on the server.

kubectl create deployment <deployment_name>

Create a new deployment.

kubectl delete deployment <deployment_name>

Delete deployments.

kubectl rollout status deployment <deployment_name>

See the rollout status of a deployment.

kubectl set image deployment/<deployment name> <container name>=image:<new image version>

Perform a rolling update (K8S default), set the image of the container to a new version for a particular deployment.

kubectl rollout undo deployment/<deployment name>

Rollback a previous deployment.

kubectl replace --force -f <configuration file>

Perform a replace deployment — Force replace, delete and then re-create the resource.

Events

kubectl get events

List recent events for all resources in the system.

kubectl get events --field-selector type=warning

List Warnings only.

kubectl get events --sort-by=.metadata.creationtimestamp

List events sorted by timestamp.

kubectl get events --field-selector involvedobject.kind=pod

List events but exclude Pod events.

kubectl get events --field-selector involvedobject.kind=node,involvedobject.name=<node_name>

Pull events for a single node with a specific name.

kubectl get events --field-selector type=normal

Filter out normal events from a list of events.

Logs

kubectl logs <pod_name>

Print the logs for a pod.

kubectl logs --since=6h <pod_name>

Print the logs for the last 6 hours for a pod.

kubectl logs --tail=50 <pod_name>

Get the most recent 50 lines of logs.

kubectl logs -f <service_name> [-c <\$container>]

Get logs from a service and optionally select which container.

kubectl logs -f <pod_name>

Print the logs for a pod and follow new logs.

kubectl logs -c <container_name> <pod_name>

Print the logs for a container in a pod.

kubectl logs <pod_name> pod.log

Output the logs for a pod into a file named 'pod.log'.

kubectl logs --previous <pod_name>

View the logs for a previously failed pod.

Namespaces

kubectl create namespace <namespace_name>

Create a namespace.

kubectl get namespace <namespace_name>

List one or more namespaces.

kubectl describe namespace <namespace_name>

Display the detailed state of one or more namespaces.

kubectl delete namespace <namespace_name>

Delete a namespace.

kubectl edit namespace <namespace_name>

Edit and update the definition of a namespace.

kubectl top namespace <namespace_name>

Display Resource (CPU/Memory/Storage) usage for a namespace.

Nodes

kubectl taint node <node_name>

Update the taints on one or more nodes.

kubectl get node

List one or more nodes.

kubectl delete node <node_name>

Delete a node or multiple nodes.

kubectl top node <node_name>

Display Resource usage (CPU/Memory/Storage) for nodes.

kubectl get pods -o wide | grep <node_name>

Pods running on a node.

kubectl annotate node <node_name>

Annotate a node.

kubectl cordon node <node_name>

Mark a node as unschedulable.

kubectl uncordon node <node_name>

Mark node as schedulable.

kubectl drain node <node_name>

Drain a node in preparation for maintenance.

kubectl label node

Add or update the labels of one or more nodes.

Pods

kubectl get pod

List one or more pods.

kubectl get pods --sort-by='.status.containerstatuses[0].restartcount'

List pods Sorted by Restart Count.

kubectl get pods --field-selector=status.phase=running

Get all running pods in the namespace.

kubectl delete pod <pod_name>

Delete a pod.

kubectl describe pod <pod_name>

Display the detailed state of a pods.

kubectl create pod <pod_name>

Create a pod.

kubectl exec <pod_name> --c <container_name> <command>

Execute a command against a container in a pod.

kubectl exec -it <pod_name> /bin/sh

Get an interactive shell on a single-container pod.

kubectl top pod

Display Resource usage (CPU/Memory/Storage) for pods.

kubectl annotate pod <pod_name> <annotations>

Add or update the annotations of a pod.

kubectl label pods <pod_name> new-label=<label name>

Add or update the label of a pod.

kubectl get pods --show-labels

Get pods and show labels.

kubectl port-forward <pod name> <port number to listen on>:<port number to forward to>

Listen on a port on the local machine and forward to a port on a specified pod.

Replication Controllers

kubectl get rc

List the replication controllers.

kubectl get rc --namespace="<namespace_name>"

List the replication controllers by namespace.

ReplicaSets

kubectl get replicaset

List ReplicaSets.

kubectl describe replicaset <replicaset_name>

Display the detailed state of one or more ReplicaSets.

kubectl scale --replicas=[x]

Scale a ReplicaSet.

Secrets

kubectl create secret

Create a secret.

kubectl get secrets

List secrets.

kubectl describe secrets

List details about secrets.

kubectl delete secret <secret_name>

Delete a secret.

Services

kubectl get services

List one or more services.

kubectl describe services

Display the detailed state of a service.

kubectl expose deployment [deployment_name]

Expose a replication controller, service, deployment, or pod as a new Kubernetes service.

kubectl edit services

Edit and update the definition of one or more services.

Services Accounts

kubectl get serviceaccounts

List service accounts.

kubectl describe serviceaccounts

Display the detailed state of one or more service accounts.

kubectl replace serviceaccount

Replace a service account.

kubectl delete serviceaccount <service_account_name>

Delete a service account.

StatefulSet

kubectl get statefulset

List StatefulSet

kubectl delete statefulset/[stateful_set_name] --cascade=false

Delete StatefulSet only (not pods).